4) Artificial Neural Network using Backpropaga

Aim:

To build an artificial neural network using backpropagation algorithm and test it with a suitable dataset.

Procedure:

→ import required python libraries.
→ Load training dataset
→ Divide data into Training, testing sets.
→ Build neural network model.
→ Predict output and calculate accuracy.

Program:

```
from sklearn.datasets import load_iris
from sklearn.neural_network import
MLPclassifier
iris = load_iris()
x,y = iris.data, iris.target
model = MLPclassifier(hidden_layer_sizes=
        (10,1), max_item = 1000, random_state=
model.fit(x,y)
Print("Predicted class: ", model,
      Predict([x[0]])[0])
Print("Accuracy: ", model.score(x,y)*100, "%")
```

Output: Predicted class: 0
Accuracy: 100.0%

Result

Output:

Predicted class : 0

Accuracy : 100 %

Result:

The artificial Neural network was trained using backpropagation tested on the dataset.

Conclusion:

Thus, the back propagation algorithm was successfully implemented to train an artificial neural Network and predict flower classes